

Forming and Solving Equations

Turn the words or the diagram into one equation, solve it, then answer the question that was asked.



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THE TOPIC ON ONE MAP

Forming and Solving Equations

01 Words into an equation

three tins: total 137 ages “think of a number” ratio of savings

02 Angles

triangle = 180° parallelogram is AC a diameter?

03 Perimeter and area (linear)

perimeter = 32 same perimeter tiles, floor plans

04 It becomes a quadratic

area = 351: “show that” Pythagoras volume, probability

05 Two unknowns

tickets, teas and coffees boys and girls

WHICH EQUATION? Find what is **equal**:

ZONE A A total

add the parts =
total

ZONE B Perimeter

add every side

ZONE C Angles

180° , 360° , 90°

ZONE D Area

multiply $\rightarrow$ often
 x^2

01 Words into an Equation

Give the smallest unknown the letter x . Write every other amount in terms of x . Then find the sentence that says what they are **equal** to.

WHEN YOU SEE → YOU DO

“4 times the number ...”, “7 fewer than ...”

→ $4x$, $x - 7$

“the total is 137”

→ add all the expressions = 137

“twice as old”, “3 years younger”

→ $2y$, $y - 3$

“work out the number of buttons in tin C”

→ solve, then **substitute** — x is not the answer

Choose x

Expressions

Equation

Solve

Answer the question

WORKED MODEL · JANUARY 2020 P2HR

Tin A: x buttons. B has 4 times as many. C has 7 fewer than A. Total 137. Find tin C.

1 $A = x$, $B = 4x$, $C = x - 7$

expressions

2 $x + 4x + x - 7 = 137$

total

3 $6x = 144 \Rightarrow x = 24$

solve

ANSWER $C = 24 - 7 = 17$ buttons

× **TRAP** Stopping at $x = 24$. Read the last line of the question again — it asked for tin C.

IDEA 01

Practise: Words into an Equation

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Think of a number

CH 2 • TOPIC 13 2 marks

I think of a number, multiply it by 3 and subtract 7. The answer is 20. Find the number.

Q2 **DR ESLAM** Ages

CH 2 • TOPIC 13 3 marks

Sam is 4 years older than Ben. Their ages add to 30. How old is Sam?

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 13

3 marks

- 2 Kalinda buys x packs of currant buns and y boxes of iced buns.

There are 6 currant buns in a pack of currant buns.

There are 8 iced buns in a box of iced buns.

Kalinda buys a total of T buns.

Write down a formula for T in terms of x and y .

(3 marks)

Q4**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 5 marks

1 Julie and Liam write down the same number.

Julie multiplies the number by 5 and then adds 4 to the result. She writes down her answer.

Liam subtracts the number from 10

He writes down his answer.

Julie's answer is two thirds of Liam's answer.

Work out the number that Julie and Liam started with.

You must show your working.

(5 marks)

Q5**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

3 marks

3 Becky has some marbles.

Chris has two times as many marbles as Becky.

Dan has seven more marbles than Chris.

They have a total of 57 marbles.

Dan says,

"If I give some marbles to Becky, each of us will have the same number of marbles."

Is Dan correct?

You must show how you get your answer.

(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 4 marks

17 Sundip and Emma have some money.

The ratio of Sundip's money to Emma's money is 3 : 5.

Emma spends £450 of her money.

The ratio of Sundip's money to Emma's money is now 2 : 3.

Find how much money Sundip has.

£

(4 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q3

CH 2 · TOPIC 13**4 marks**

- 3 Three tins, A , B and C , each contain buttons.

Tin A contains x buttons.

Tin B contains 4 times the number of buttons that tin A contains.

Tin C contains 7 fewer buttons than tin A .

The total number of buttons in the three tins is 137

Work out the number of buttons in tin C .

(Total for Question 3 is 4 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q2

CH 2 · TOPIC 13**6 marks**

- 2 (a) Factorise fully $12pq - 18p$

There are 56 metal bars in a box.
Each metal bar is gold or silver or zinc.

y metal bars are gold.
 $(3y + 7)$ metal bars are silver.
 $(2y - 5)$ metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.
Show clear algebraic working.

(Total for Question 2 is 6 marks)

02 Angles

Angles in a diagram with x in them: find the angle fact that links them, write the equation, solve.

WHEN YOU SEE → YOU DO

a triangle

→ angles add to 180°

around a point / a quadrilateral

→ 360°

a right angle, a right-angled triangle

→ the other two add to 90°

parallel lines, a parallelogram

→ co-interior add to 180° ; opposite angles equal

“is AC a diameter?”

→ show the angle = 180°

WORKED MODEL · TRIANGLE

A triangle has angles 30° , $(4x + 10)^\circ$ and $(x + 20)^\circ$. Find x .

1 $30 + 4x + 10 + x + 20 = 180$

triangle = 180°

2 $5x + 60 = 180 \Rightarrow 5x = 120$

collect

ANSWER $x = 24$

ANSWER THE QUESTION “the smallest angle”, “the largest angle”: substitute x into every angle, then choose.

IDEA 02

Practise: Angles

My questions first, then selected practice.

Q9**DR ESLAM** Triangle

CH 2 · TOPIC 13 2 marks

The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find the largest angle.

SELECTED PRACTICE

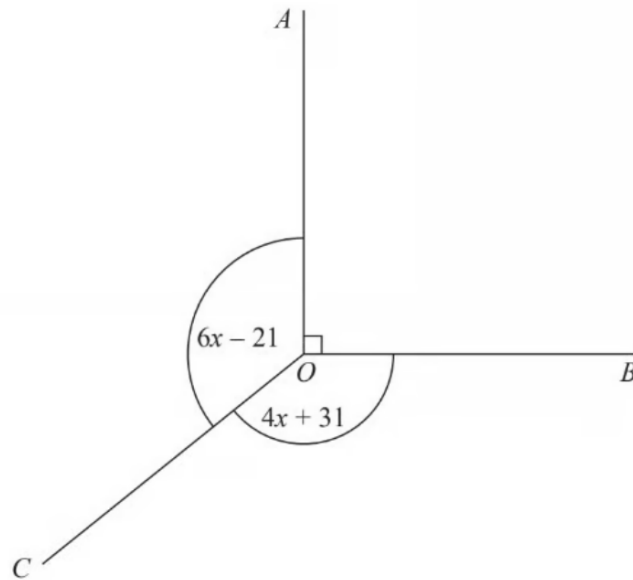
Q10

PRACTICE

Practice · Easy

CH 2 · TOPIC 13 3 marks

4

Diagram **NOT**
accurately drawn

In the diagram, all angles are in degrees.

Angle AOB is a right angle.

Angle $AOC = \text{Angle } BOC$.

Work out the value of x .

(3 marks)

Q11**PRACTICE**

Practice · Easy

CH 2 · TOPIC 13 4 marks

6 The diagram shows a triangle.

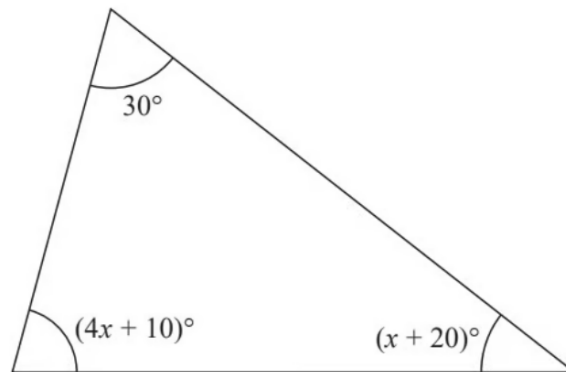


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

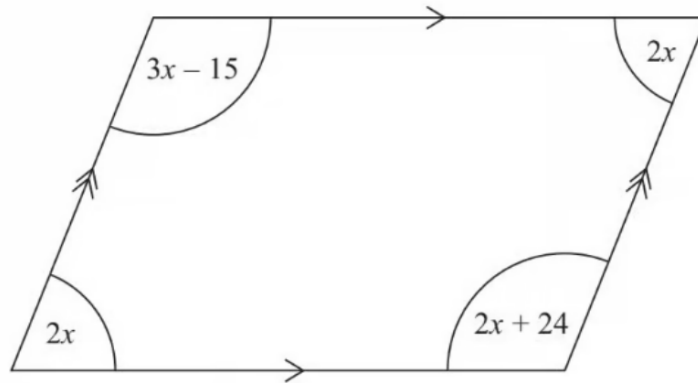
(4 marks)

Q12**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 3 marks

7

Diagram **NOT**
accurately drawn

The diagram shows a parallelogram.

The sizes of the angles, in degrees, are

$$2x$$

$$3x - 15$$

$$2x$$

$$2x + 24$$

Work out the value of x .

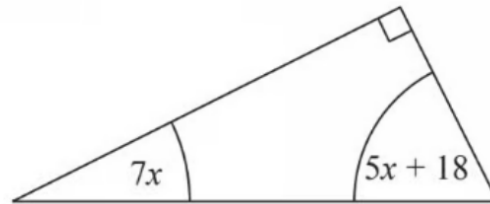
(3 marks)

Q13**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 **3 marks**

9 The diagram shows a right-angled triangle.



All the angles are in degrees.

Work out the size of the smallest angle of the triangle.

(3 marks)

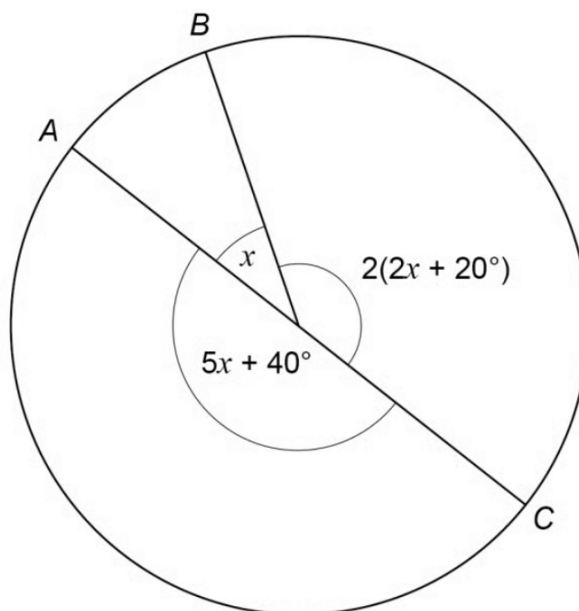
Q14 PRACTICE Practice · Medium

CH 2 · TOPIC 13 3 marks

15 A , B and C are three points on a circle.

The radii from A , B and C are shown.

Not drawn
accurately



Is AC a diameter of the circle?
You **must** show your working.

(3 marks)

Q15**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

16 $ABCD$ is a trapezium.

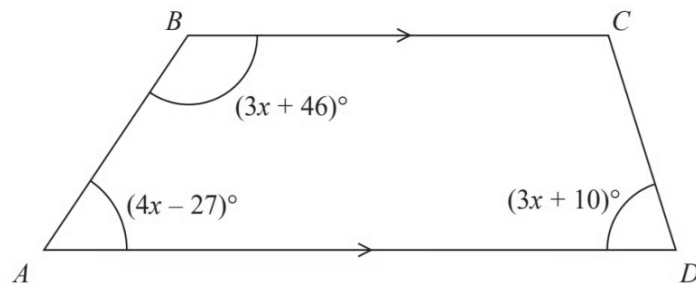


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

(4 marks)

03 Perimeter and Area (Linear)

Perimeter: add every side — including the sides with no label. Two shapes with the same perimeter or area: write both, make them equal.

WHEN YOU SEE → YOU DO

“the perimeter is 32”

→ add **all** sides = 32

a rectangle with $2x + 6$ and $5x - 9$ on opposite sides

→ opposite sides equal

“the two shapes have the same perimeter”

→ perimeter A = perimeter B

tiles, floor plans

→ find the missing lengths first

Label every side

Add or multiply

Equation

Solve

WORKED MODEL · NOVEMBER 2021 P2H

Rectangle A: 4 by $(5 - x)$. Rectangle B: 5 by $(2x - 1)$. Area B is twice area A. Find x .

1 $5(2x - 1) = 2 \times 4(5 - x)$

area B = $2 \times$ area A

2 $10x - 5 = 40 - 8x$

expand

3 $18x = 45$

collect

ANSWER $x = 2.5$

IDEA 03

Practise: Perimeter and Area (Linear)

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Perimeter

CH 2 • TOPIC 13 3 marks

A rectangle is $(x + 3)$ cm by $(2x - 1)$ cm. Its perimeter is 34 cm. Work out its area.

SELECTED PRACTICE

Q17 PRACTICE Practice · Easy

CH 2 · TOPIC 13 3 marks

- 1 The diagram shows a rectangle.

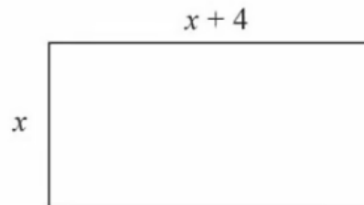


Diagram **NOT**
accurately drawn

All measurements are given in centimetres.

The perimeter of the rectangle is 45 cm.

Work out the value of x .

(3 marks)

Q18 PRACTICE Practice · Medium

CH 2 · TOPIC 13 5 marks

10 ABC is a triangle.

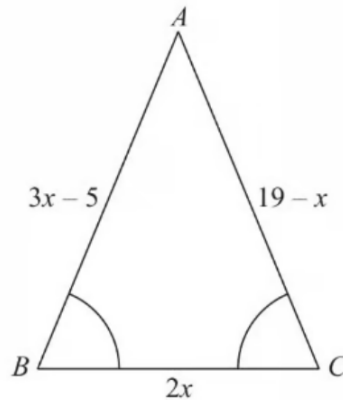


Diagram **NOT**
accurately drawn

Angle $ABC =$ angle BCA .

The length of side AB is $(3x - 5)$ cm.

The length of side AC is $(19 - x)$ cm.

The length of side BC is $2x$ cm.

Work out the perimeter of the triangle.

Give your answer as a number of centimetres.

(5 marks)

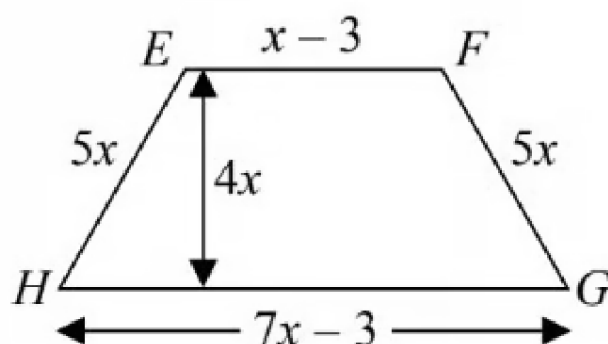
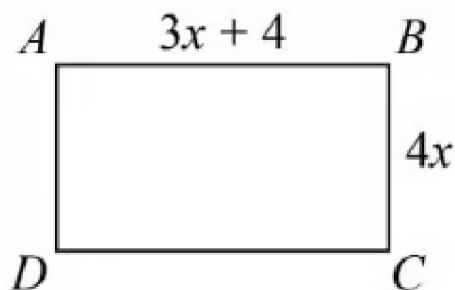
Q19**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 5 marks

5 $ABCD$ is a rectangle.

$EFGH$ is a trapezium.



All measurements are in centimetres.

The perimeters of these two shapes are the same.

Work out the area of the rectangle.

(5 marks)

Q20**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

5 marks

18 The diagram shows a hexagon $ABCDEF$

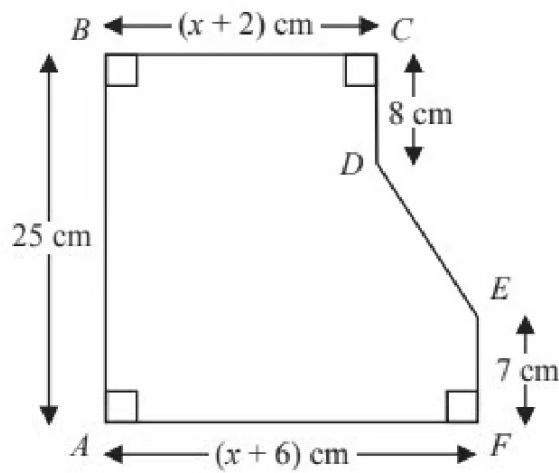


Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon $ABCDEF$ is 258 cm^2

Work out the value of x

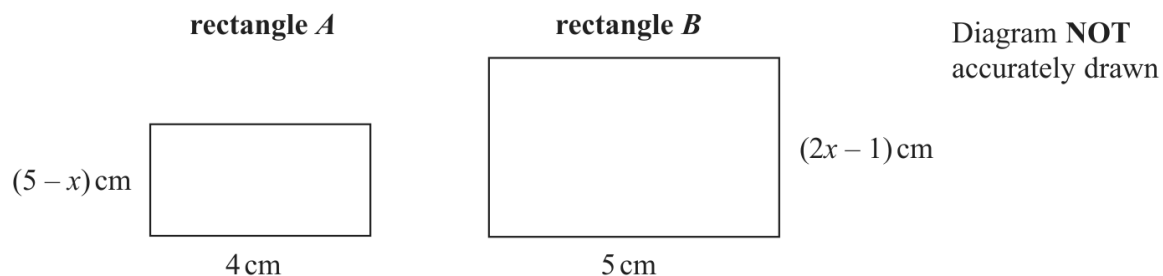
(5 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 2H · Q2

CH 2 · TOPIC 13 4 marks

- 2 Here are two rectangles, rectangle A and rectangle B .



The area of rectangle B is twice the area of rectangle A .

Work out the value of x .
Show your working clearly.

$x =$

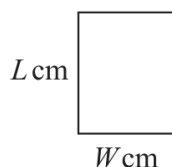
(Total for Question 2 is 4 marks)

Q22**PAST PAPER**

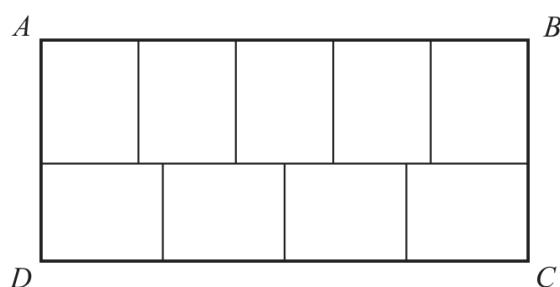
Edexcel 4MA1 · November 2021 · Paper 1H · Q5

CH 2 · TOPIC 13 5 marks

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

(Total for Question 5 is 5 marks)

04 It Becomes a Quadratic

Multiply two lengths with x in both and you get x^2 . ERCS, solve — then **reject** the answer that makes a length negative.

WHEN YOU SEE → YOU DO

area of a shape with x in two sides

→ a quadratic: ERCS, then solve

“Show that $2x^2 + x - 351 = 0$ ”

→ start from the formula, end at exactly that line

a right-angled triangle with x sides

→ Pythagoras: $a^2 + b^2 = c^2$

two answers, one negative

→ a length cannot be negative: say why you reject it

Formula

Expand

= 0

Solve

Reject

WORKED MODEL · NOVEMBER 2025 P2H

$ax^2 + 4x + c = 0$ has solutions $x = \frac{-4 \pm 2\sqrt{39}}{10}$. Find a and c .

1 $2a = 10 \Rightarrow a = 5$

compare the bottom

2 $\sqrt{16 - 4ac} = 2\sqrt{39} = \sqrt{156}$

compare the root

3 $16 - 20c = 156 \Rightarrow c = -7$

solve

ANSWER $a = 5, c = -7$

× **TRAP** In Kate's box question, $x = -40$ is a width: it is impossible. Only $x = 35$.

IDEA 04

Practise: It Becomes a Quadratic

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Area to quadratic

CH 2 · TOPIC 13 4 marks

A rectangle is x cm wide and $(x + 3)$ cm long. Its area is 40 cm^2 . Find x .

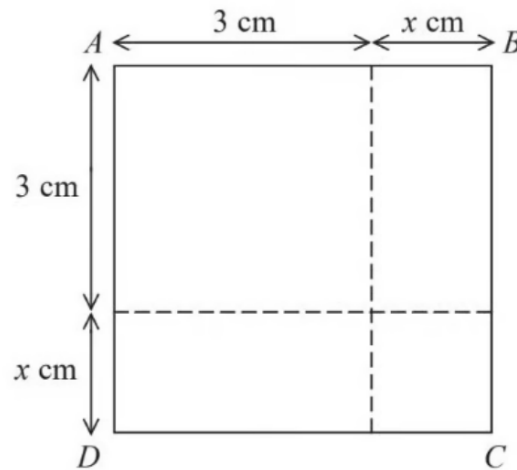
$x =$ _____

SELECTED PRACTICE

Q24 PRACTICE Practice · Easy

CH 2 · TOPIC 13 3 marks

5



The area of square $ABCD$ is 10 cm^2 .

Show that $x^2 + 6x = 1$

(3 marks)

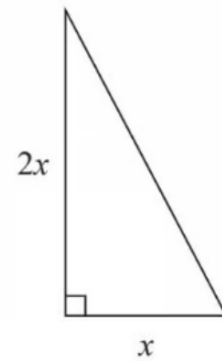
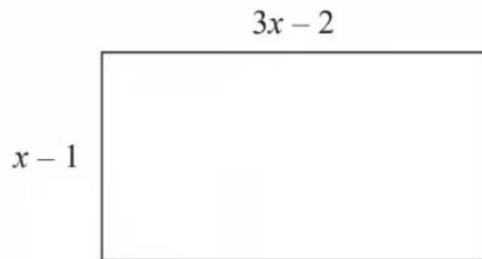
Q25**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13

5 marks

6 Here is a rectangle and a right-angled triangle.



All measurements are in centimetres.

The area of the rectangle is greater than the area of triangle.

Find the set of possible values of x .

(5 marks)

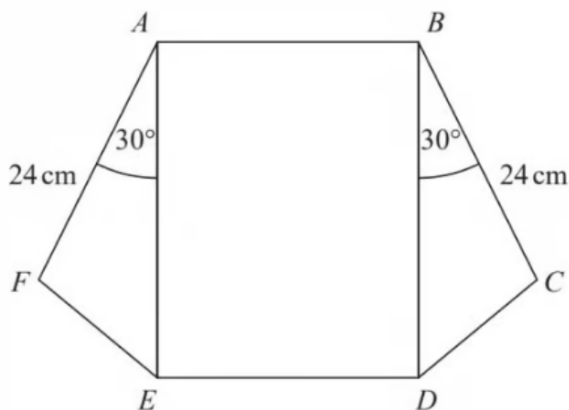
Q26**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13

4 marks

- 5 The diagram shows a rectangle, $ABDE$, and two congruent triangles, AFE and BCD .



area of rectangle $ABDE$ = area of triangle AFE + area of triangle BCD

$$AB : AE = 1 : 3$$

Work out the length of AE .

.....cm

(4 marks)

Q27**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13 4 marks

11 A sphere has radius $2x$ cm

A cone has

radius $3x$ cm

perpendicular height h cm

The sphere and the cone have the same volume.

Work out radius of cone : perpendicular height of cone

Give your answer in the form $a : b$ where a and b are integers.

(4 marks)

PAST PAPERS

Q28**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q20

CH 2 · TOPIC 13 **3 marks**

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c

Show your working clearly.

$a =$

$c =$

(Total for Question 20 is 3 marks)

05 Two Unknowns

Two unknown amounts and two facts? Use two letters, write two equations, and solve them simultaneously (Topic 12).

WHEN YOU SEE → YOU DO

“3 adult tickets and 1 child ticket cost £30”

→ $3a + c = 30$

“if 5 more boys and 8 more girls join ...”

→ write the new amounts, then the fact

“half as many boys as girls”

→ $2(\text{boys}) = \text{girls}$

WORKED MODEL · TICKETS

3 adults and 1 child cost £30. 1 adult and 3 children cost £22. Find each price.

1 $3a + c = 30, a + 3c = 22$

two letters, two facts

2 $9a + 3c = 90 \Rightarrow 8a = 68$

×3, subtract

ANSWER adult £8.50, child £4.50

WHERE YOU WILL USE IT Every problem-solving question on the paper — geometry, probability, ratio — ends with “form an equation and solve it”.

IDEA 05

Practise: Two Unknowns

My questions first, then selected practice.

Q29**DR ESLAM**

Pens and pencils

CH 2 • TOPIC 13

4 marks

3 pens and 2 pencils cost £2.60. 1 pen and 2 pencils cost £1.40. Find the cost of one pen.

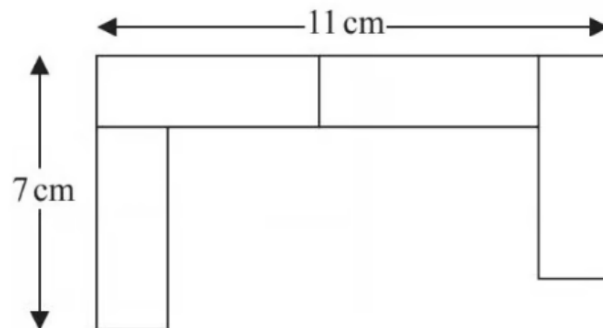
SELECTED PRACTICE

Q30**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

- 2 A pattern is made using identical rectangular tiles.



Find the total area of the pattern.

(4 marks)

Q31**PRACTICE**

Practice · Medium

CH 2 · TOPIC 13 4 marks

- 6 3 kg of potatoes and 4 kg of carrots have a total cost of 440p.
4 kg of potatoes and 3 kg of carrots have a total cost of 470p.

Work out the total cost of 1 kg of potatoes and 1 kg of carrots.

(4 marks)**Q32****PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 4 marks

- 7 A cinema sells adult tickets and child tickets.

The total cost of 3 adult tickets and 1 child ticket is £30

The total cost of 1 adult ticket and 3 child tickets is £22

Work out the cost of an adult ticket and the cost of a child ticket.

(4 marks)

Q33**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 4 marks

- 8 3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

(4 marks)

Q34**PRACTICE**

Practice · Hard

CH 2 · TOPIC 13 5 marks

13 (a) In a chess club, there are x boys and y girls.

If 5 more boys and 8 more girls join, there would be half as many boys as girls.

Show that $y = 2x + 2$

(2 marks)

(b) If instead,

10 more boys and 1 more girl join, there would be the same number of boys and girls.

Work out x and y .

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(3 marks)**

Q35**PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 13 5 marks

4 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{6}{13}$

Find the number of red counters and the number of green counters that were in the bag originally.

(5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

Choose x

Expressions

Equation

Solve

Answer the question

- | | | |
|-----------|-------------------------|--|
| 01 | Words | Smallest unknown = x ; add the parts = total; answer what was asked. |
| 02 | Angles | 180° triangle, 360° point/quadrilateral, parallel lines. |
| 03 | Perimeter / area | Add every side; same perimeter $\rightarrow$ make them equal. |
| 04 | Quadratic | $x \times x$ gives x^2 : ERCS, solve, reject negative lengths. |
| 05 | Two unknowns | Two letters, two equations, eliminate. |

BOTTOM LINE What is **equal** to what? $\rightarrow$ write it $\rightarrow$ solve $\rightarrow$ substitute back and answer the question.

Answers

Forming and Solving Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • WORDS INTO AN EQUATION

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam • Think of a number | $3x - 7 = 20; x = 9$ |
| Q2 | Dr Eslam • Ages | $x + x + 4 = 30; \text{ Ben 13, Sam 17}$ |
| Q3 | Practice • Easy | $T = 6x + 8y$ |
| Q4 | Practice • Medium | $5n + 4 = \frac{2}{3}(10 - n); n = \frac{8}{17}$ |
| Q5 | Practice • Hard | Becky 10, Chris 20, Dan 27: equal shares need 19 each, but Chris has 20 — No |
| Q6 | Practice • Hard | £2700 |
| Q7 | Edexcel 4MA1 • January 2020 • Paper 2HR • Q3 | $6x - 7 = 137, x = 24: 17 \text{ buttons}$ |
| Q8 | Edexcel 4MA1 • November 2023 • Paper 1H • Q2 | (a) $6p(2q - 3)$ (b) $6y + 2 = 56, y = 9: 13 \text{ zinc bars}$ |

IDEA 02 • ANGLES

- Q9** Dr Eslam · Triangle $6x = 180, x = 30: 90^\circ$
- Q10** Practice · Easy $6x - 21 = 4x + 31; x = 26$
- Q11** Practice · Easy $x = 24$
- Q12** Practice · Medium $x = 39$
- Q13** Practice · Medium $12x + 18 = 90, x = 6: 42^\circ$
- Q14** Practice · Medium $x = 28: 5x + 40 = 180^\circ, \text{ so yes}$
- Q15** Practice · Medium $x = 23: \text{ largest angle } 115^\circ$

IDEA 03 · PERIMETER AND AREA (LINEAR)

- Q16** Dr Eslam · Perimeter $6x + 4 = 34, x = 5: 8 \times 9 = 72 \text{ cm}^2$
- Q17** Practice · Easy $4x + 8 = 45; x = 9.25$
- Q18** Practice · Medium $x = 6: 13 + 13 + 12 = 38 \text{ cm}$
- Q19** Practice · Hard $x = 3.5: 203 \text{ cm}^2$
- Q20** Practice · Hard $25(x + 6) - 52 = 258; x = 6.4$
- Q21** Edexcel 4MA1 · November 2021
· Paper 2H · Q2 $x = 2.5$
- Q22** Edexcel 4MA1 · November 2021
· Paper 1H · Q5 $5W = 4L, 9LW = 1620: L = 15, W = 12$

IDEA 04 · IT BECOMES A QUADRATIC

- Q23** Dr Eslam · Area to quadratic $x^2 + 3x - 40 = 0, (x+8)(x-5) = 0: x = 5$ ($x = -8$ rejected)
- Q24** Practice · Easy $(x+3)^2 = 10 \Rightarrow x^2 + 6x = 1$
- Q25** Practice · Hard $2x^2 - 5x + 2 > 0$ and $x > 1: x > 2$
- Q26** Practice · Very Hard $3x^2 = 36x, x = 12: AE = 36 \text{ cm}$
- Q27** Practice · Very Hard $h = \frac{32}{9}x: 27 : 32$
- Q28** Edexcel 4MA1 · November 2025
· Paper 2H · Q20 $a = 5, c = -7$

IDEA 05 · TWO UNKNOWNNS

- Q29** Dr Eslam · Pens and pencils $2p = 1.20: \text{pen } 60p, \text{pencil } 40p$
- Q30** Practice · Medium $L = 4, W = 3: 48 \text{ cm}^2$
- Q31** Practice · Medium £1.30
- Q32** Practice · Hard adult £8.50, child £4.50
- Q33** Practice · Hard tea £1.40, coffee £1.80
- Q34** Practice · Hard (a) $2(x+5) = y+8$ (b) $x = 7, y = 16$
- Q35** Practice · Very Hard 12 red, 9 green